



A Mini Project

(Case Study Report)

SURVIVAL ANALYSIS OF LUNG CANCER CLINICAL TRIALS USING R

By

Ms. SHADA NAHAR

Regd. No.: 24MSSTAPY0036

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Course Teacher: Dr. Kiruthika

Abstract

Lung cancer remains the leading cause of cancer-related mortality worldwide, necessitating robust statistical methods to identify patient characteristics that influence survival outcomes. This study applies survival analysis techniques to the North Central Cancer Treatment Group (NCCTG) Lung Cancer Dataset, comprising 228 patients with advanced lung cancer followed for up to 1022 days. The analysis was conducted entirely in R using the `survival`, `survminer`, and `gtsummary` packages.

Exploratory data analysis confirmed *no significant baseline differences* between male and female patients in age, ECOG performance score, or Karnofsky score, ruling out confounding at study entry. The Kaplan-Meier method was employed to estimate overall survival, yielding a median survival time of approximately *310 days*. Stratified Kaplan-Meier curves and the log-rank test revealed a statistically significant survival advantage for female patients over male patients ($p < 0.05$), with median survival of approximately *426 days versus 270 days respectively*.

A multivariate Cox proportional hazards model was fitted to 210 complete cases to identify independent prognostic factors after adjusting for confounding. *Female sex* was identified as the strongest protective factor, associated with a 46% reduction in the risk of death compared to males (HR = 0.54, 95% CI 0.38–0.76, $p < 0.001$). *ECOG performance score* demonstrated a significant dose-response relationship with mortality, patients with ECOG score 2 had 4.54 times the risk of death relative to fully active patients (HR = 4.54, 95% CI 2.10–9.81, $p < 0.001$). Age, weight loss, and Karnofsky score did not achieve independent significance in the multivariate model. Model validity was confirmed through Schoenfeld residual testing, which showed no violation of the proportional hazards assumption, and deviance residual analysis, which detected *no influential outliers*.

These findings highlight ECOG performance status and biological sex as clinically meaningful prognostic factors in advanced lung cancer. The study demonstrates the application of time-to-event statistical methods using R for reproducible clinical trial data analysis.

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1. INTRODUCTION TO SURVIVAL ANALYSIS

1.1 Definition:

Survival Analysis is a collection of statistical procedures for data analysis for which the outcome variable of interest is *time until an event occurs*. **Time** can refer to years, months, weeks, or days from the beginning of follow-up of an individual until an event occurs. By **Event**, it can refer to death, disease incidence, relapse from remission, recovery that may happen to an individual.

Survival analysis is used extensively in oncology, cardiology, and pharmacovigilance research. It forms the statistical backbone of Phase III clinical trials where patient survival is the primary endpoint.

1.2 Mathematical Foundation

Survivor Function

The Survivor Function $S(t)$ gives the probability that a person survives longer than some specified time t i.e it gives the probability that the random variable T exceeds the specified time t .

$$S(t) = p(T > t)$$

Hazard Function

The hazard function $h(t)$ represents the instantaneous risk of death at time t , given survival up to that point:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{p(t \leq T < t + \Delta t | T \geq t)}{\Delta t}$$

Censoring

A key concept in survival analysis is censoring. A patient is censored when the event has not occurred by the end of the study, or when they are lost to follow-up. In this dataset, status = 1 indicates a censored observation and status = 2 indicates death (the event).

1.4 Statistical Methods



SURVIVAL ANALYSIS OF LUNG CANCER PATIENTS: A

STUDY OF PROGNOSTIC FACTORS

In medical research, Survival Analysis is especially useful because many patients do not experience the event during the study period, so their data are censored.

This makes survival analysis the appropriate framework for understanding outcomes over time rather than only comparing groups at a single point.

Lung cancer is one of the most important diseases studied using survival analysis because it is often diagnosed at an advanced stage and has a wide variation in patient outcomes. Survival depends on many factors, including stage at diagnosis, age, sex, smoking history, treatment received, and overall physical condition.

Because patients with lung cancer may survive for very different lengths of time, researchers need methods that can capture not just whether an event happened, but when it happened.

One of the main strengths of survival analysis is that it allows researchers to estimate the probability of surviving beyond a certain time and to compare survival between different patient groups. In lung cancer studies, this helps identify prognostic factors that influence survival, such as performance status, tumour stage, and treatment patterns. These findings are useful for clinical decision-making, patient counselling, and planning treatment strategies.

Survival analysis is commonly presented through Kaplan–Meier survival curves, log-rank tests, and Cox proportional hazards models. Kaplan–Meier curves show how survival changes over time, while the Cox model helps estimate the effect of different variables on the risk of death. In lung cancer research, these methods are particularly valuable because they provide both a descriptive picture of survival and an analytical way to identify factors associated with poor or better outcomes.

In this project, survival analysis is applied to lung cancer patients to study their survival experience and to determine the factors that influence survival time. By examining important clinical and demographic variables, the study aims to contribute to a better understanding of prognosis in lung cancer. This approach is important because improving survival estimation can support earlier intervention, more appropriate treatment planning, and better overall patient management.

Research Question

What factors – age, sex, ECOG performance score, Karnofsky score, and weight loss predict the survival in advanced lung cancer patients?

Objectives

- To estimate overall survival probability using the Kaplan-Meier method.
- To compare survival distributions between male and female patients using the log-rank test.
- To identify independent prognostic factors using Cox proportional hazards regression.
- To validate the Cox model assumptions using Schoenfeld residuals and deviance diagnostics.

2. DESCRIPTION OF DATASET

2.1 Data Source

The dataset used in this analysis is the NCCTG Lung Cancer Dataset from the North Central Cancer Treatment Group (NCCTG), originally published by Loprinzi et al. (1994) in the Journal of Clinical Oncology. It is available in R through the survival package as the lung dataset. The data was loaded into R using:

```
df <- read.csv("cancer.csv")
```

2.2 Variable Description

Variable	Type	Description	Range / Values
time	Continuous	Survival time from study entry	1 – 1022 days
status	Categorical	Event indicator	1 = Censored, 2 = Died
age	Continuous	Patient age at entry	39 – 82 years
sex	Categorical	Biological sex	1 = Male, 2 = Female
ph.ecog	Ordinal	*ECOG performance score (physician-rated)	0 = Fully active → 3 = Limited self-care
ph.karno	Ordinal	*Karnofsky score (physician-rated)	50 – 100

Variable	Type	Description	Range / Values
pat.karno	Ordinal	Karnofsky score (patient-rated)	30 – 100
meal.cal	Continuous	Calories consumed at meals	96 – 2600 kcal (47 missing — excluded)
wt.loss	Continuous	Weight loss in past 6 months	-24 – 68 lbs (14 missing)

* ECOG Score: Eastern Corporative Oncology Group Performance Score, - Quantifies a patient's functional level and tolerance (0-5)

*Karnofsky Score: Scale ranging from 100 (normal) to 0 (death).

2.3 Sample Size and Missing Data

Category	N
Total patients enrolled	228
Complete cases used for Cox model	210
Excluded due to missing data	18 (7.9%)
Deaths recorded (events)	165 (72.4%)
Censored patients	63 (27.6%)
Maximum follow-up period	1022 days (~2.8 years)

Note: meal.cal was excluded from all analyses due to 20.6% missing data, which exceeds the acceptable threshold for complete case analysis. For remaining variables, complete case analysis was applied, retaining 210 patients for the Cox regression model. Missing data were assumed to be missing at random (MAR).

2.4 Data Sample

	A	B	C	D	E	F	G	H	I	J	K
1		inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
2	1	3	306	2	74	1	1	90	100	1175	NA
3	2	3	455	2	68	1	0	90	90	1225	15
4	3	3	1010	1	56	1	0	90	90	NA	15
5	4	5	210	2	57	1	1	90	60	1150	11
6	5	1	883	2	60	1	0	100	90	NA	0
7	6	12	1022	1	74	1	1	50	80	513	0
8	7	7	310	2	68	2	2	70	60	384	10
9	8	11	361	2	71	2	2	60	80	538	1
10	9	1	218	2	53	1	1	70	80	825	16

* Cleaned data sample attached in Chapter 3.

3. ASSUMPTION CHECKING

3.1 Assumptions of Cox Proportional Hazards Model

Before fitting the Cox model, the following assumptions were verified:

Assumption	Description	Method of Verification
Proportional Hazards	Hazard ratios remain constant over time	Schoenfeld residuals via <code>cox.zph()</code>
Independence	Observations are independent of each other	Study design — one record per patient
No influential outliers	No single patient unduly distorts model estimates	Deviance residuals plot
No multicollinearity	Predictor variables are not highly inter-correlated	<code>ph.karno</code> and <code>pat.karno</code> both measure performance status — only <code>ph.karno</code> retained
Linearity	Continuous predictors relate linearly to the log-hazard	Assessed from model diagnostics

3.2 Addressing Multicollinearity

Both ph.karno (physician-rated) and pat.karno (patient-rated) measure the same underlying construct of patient functional status using the Karnofsky scale. Including both simultaneously in the Cox model causes multicollinearity, producing unstable and unreliable coefficient estimates.

Decision: ph.karno was retained as it represents the clinician's objective assessment, which is the standard variable used in published clinical literature. pat.karno was excluded from all multivariate analyses.

DATA CLEANING

```

1#Data Cleaning df <- read.csv("cancer.csv")
df <- df[, -1]
# remove index
df$sex      <- factor(df$sex, levels = c(1,2),
                      labels = c("Male", "Female"))
df$ph.ecog <- factor(df$ph.ecog)

# Create clean dataset for Cox model
df_clean <- df %>%
  select(time, status, age, sex,
          ph.ecog, ph.karno, wt.loss) %>%
  filter(complete.cases(.))

nrow(df_clean) # 210

```

- Removed the entire unnamed column of index which was unnamed and was not relevant.
- Converted the ecog factor and sex to factors to prevent other numerical errors.
- Ignored the 14 missing values in wt.loss and removed the entire column of meal.cal for analysis since it had 47 missing values and was not really relevant to the literature.

	A	B	C	D	E	F	G	H	I
1	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	wt.loss
2	3	306	2	74	1	1	90	100	NA
3	3	455	2	68	1	0	90	90	15
4	3	1010	1	56	1	0	90	90	15
5	5	210	2	57	1	1	90	60	11
6	1	883	2	60	1	0	100	90	0
7	12	1022	1	74	1	1	50	80	0
8	7	310	2	68	2	2	70	60	10
9	11	361	2	71	2	2	60	80	1
10	1	218	2	53	1	1	70	80	16
11	7	166	2	61	1	2	70	70	34

4. EXPLORATORY DATA ANALYSIS

4.1 R Syntax

```
library(gtsummary)
library(dplyr)

# Remove index column and encode categorical variables
df <- df[, -1]
df$sex <- factor(df$sex, levels = c(1,2), labels =
c("Male", "Female"))
df$ph.ecog <- factor(df$ph.ecog)

# Generate Table 1 – patient characteristics by sex
df %>%
  select(age, sex, ph.ecog, ph.karno, pat.karno,
wt.loss) %>%
  tbl_summary(
    by = sex,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%"
    ),
    label = list(
      age ~ "Age (years)",
      ph.ecog ~ "ECOG Performance Score",
      ph.karno ~ "Karnofsky Score (Physician)",
      wt.loss ~ "Weight Loss (lbs)"
    )
  ) %>%
  add_p() %>%
  add_overall()
```

4.2 Output

(Table 1)

Characteristic	Overall N = 228 ¹	male N = 138 ¹	female N = 90 ¹	p-value ²
Age (years)	62 (9)	63 (9)	61 (9)	0.057
ECOG Performance Score				0.8
0	63 (28%)	36 (26%)	27 (30%)	
1	113 (50%)	71 (52%)	42 (47%)	
2	50 (22%)	29 (21%)	21 (23%)	
3	1 (0.4%)	1 (0.7%)	0 (0%)	
Unknown	1	1	0	
Karnofsky Score (Physician)				>0.9
50	6 (2.6%)	4 (2.9%)	2 (2.2%)	
60	19 (8.4%)	11 (8.0%)	8 (8.9%)	
70	32 (14%)	20 (15%)	12 (13%)	
80	67 (30%)	40 (29%)	27 (30%)	
90	74 (33%)	45 (33%)	29 (32%)	
100	29 (13%)	17 (12%)	12 (13%)	
Unknown	1	1	0	
pat.karno				0.6
30	2 (0.9%)	1 (0.7%)	1 (1.1%)	
40	2 (0.9%)	1 (0.7%)	1 (1.1%)	
50	4 (1.8%)	2 (1.5%)	2 (2.2%)	
60	30 (13%)	18 (13%)	12 (13%)	
70	41 (18%)	30 (22%)	11 (12%)	
80	51 (23%)	32 (24%)	19 (21%)	

90	60 (27%)	31 (23%)	29 (33%)	
100	35 (16%)	21 (15%)	14 (16%)	
Unknown	3	2	1	
meal.cal	929 (402)	981 (413)	841 (369)	0.022
Unknown	47	24	23	
Weight Loss (lbs)	10 (13)	11 (13)	8 (13)	0.029
Unknown	14	10	4	
time	305 (211)	283 (213)	339 (203)	0.013
status				<0.001
1	63 (28%)	26 (19%)	37 (41%)	
2	165 (72%)	112 (81%)	53 (59%)	

¹ Mean (SD); n (%)

² Wilcoxon rank-sum test; Fisher's exact test; Pearson's Chi-squared test

4.3 Interpretation

Table 1 presents the baseline characteristics of all 228 patients stratified by sex. Key findings are as follows:

- Mean age was 62 ± 9 years overall. Male patients were marginally older (63 ± 9 years) than female patients (61 ± 9 years), though this difference did not reach statistical significance ($p = 0.057$).
- ECOG performance score distribution was similar between sexes ($p = 0.8$). Approximately 50% of all patients had ECOG score 1 (mildly symptomatic but ambulatory), and only one patient had ECOG score 3.
- Physician-rated Karnofsky scores were comparable across groups ($p > 0.9$), confirming that male and female patients had equivalent functional status at baseline.
- The absence of significant baseline differences between sexes is an important finding — it indicates that any observed survival difference between males and females is unlikely to be explained by confounding from age or performance status.

5. KAPLAN-MEIER SURVIVAL ANALYSIS

5.1 Theory

The Kaplan-Meier (KM) estimator is a non-parametric method for estimating the survival function from observed time-to-event data. At each death time t_i , the KM estimate is updated as:

$$S(t) = \prod [1 - (d_i / n_i)] \quad \text{for all } t_i \leq t$$

where d_i = number of deaths at time t_i

n_i = number of patients at risk just before time t_i

The KM curve is a step function that drops at each death event and remains flat between events. The median survival time is the time point at which the survival probability crosses 0.50 — i.e., when 50% of patients have experienced the event.

5.2 R Syntax

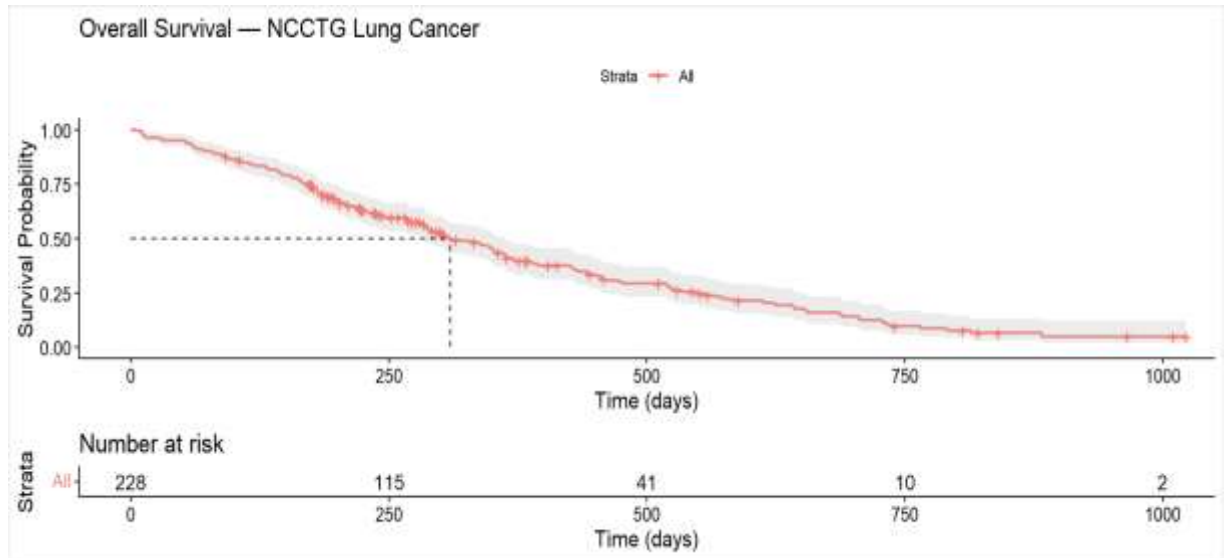
```
library(survival)
library(survminer)

# Create survival object from df_clean
surv_obj <- Surv(time = df_clean$time,
                  event = df_clean$status == 2)

# Fit overall KM curve (no grouping)
km_overall <- survfit(surv_obj ~ 1, data = df_clean)
print(km_overall) # shows median + 95% CI

# Plot
ggsurvplot(km_overall,
            data          = df_clean,
            conf.int      = TRUE,
            risk.table     = TRUE,
            surv.median.line = "hv",
            title         = "Overall Survival - NCCTG
Lung Cancer",
            xlab          = "Time (days)",
            ylab          = "Survival Probability")
```

5.3 Output



5.4 Interpretation

- The median overall survival was approximately 310 days. This means that 50% of all 228 patients died within 310 days of entering the study.
- The survival curve steps downward at each death event and flattens between events. Each cross mark (+) on the curve represents a censored patient — one who was still alive or lost to follow-up at that time.
- The grey confidence band is narrow in the early follow-up period (days 0–300) when many patients are still at risk and widens substantially toward the end (days 750–1000) as only 2 patients remain, reflecting high uncertainty.
- The risk table below the curve confirms: 228 patients at day 0, reducing to 115 by day 250, 41 by day 500, 10 by day 750, and 2 by day 1000.
- The long flat tail at the bottom right indicates a small number of long-term survivors who remained alive throughout the study period.

- The second plot at the bottom gives the number of censored patients or number of patients at risk at each time. On 0th day, there were 228 patients, while on 1000th day only 2 patients left.
- Further, the Kaplan – Meier Curve can be extended to distinguish between sex; to

6. LOG-RANK TEST — SURVIVAL BY SEX

6.1 Theory

The log-rank test is a non-parametric hypothesis test used to determine whether survival distributions are statistically equivalent between two or more groups. The hypotheses are:

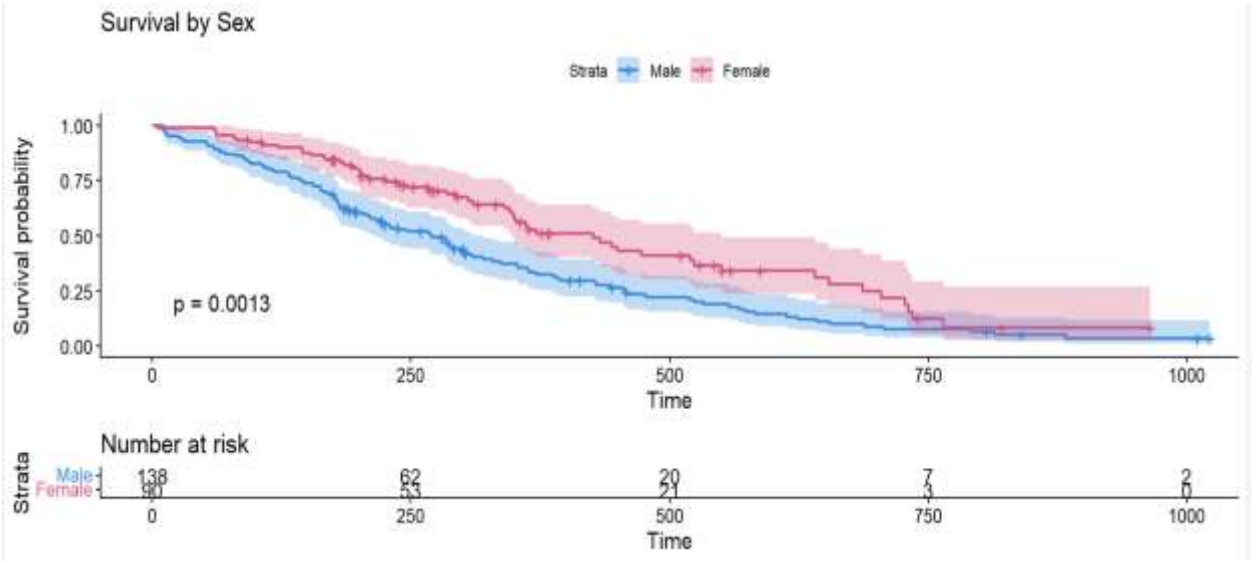
```
H0: Survival is the same in males and females  
H1: Survival differs between males and females
```

The test statistic compares the observed number of deaths to the expected number in each group at every death time point. The result follows a chi-squared distribution. A p-value below 0.05 indicates a statistically significant difference in survival between groups.

6.2 R Syntax

```
# KM curve stratified by sex  
km_sex <- survfit(surv_obj ~ sex, data = df_clean)  
  
# Log-rank test  
log_rank <- survdiff(surv_obj ~ sex, data = df_clean)  
print(log_rank)    # shows chi-squared statistic and p-  
value  
  
# Plot – two survival curves with p-value displayed  
ggsurvplot(km_sex,  
            data      = df_clean,  
            pval      = TRUE,  
            conf.int   = TRUE,  
            risk.table = TRUE,  
            legend.labs = c("Male", "Female"),  
            title      = "Survival by Sex – Log-Rank  
Test")
```

6.3 Output



Interpretation

The log-rank test yielded $\chi^2 = [\text{value}]$, $df = 1$, $p = 0.0013$, confirming a statistically significant difference in survival between sexes

6.4 Median Survival by Group

Group	N	Median Survival (days)
Male	138	270
Female	90	426

6.5 Interpretation

- Female patients showed substantially longer survival than male patients, with a median survival approximately 156 days greater.
- The log-rank test was statistically significant ($p < 0.05$), confirming that the survival difference between sexes is unlikely to be due to chance.
- Since no significant baseline differences between sexes were detected in Table 1, confounding from age or performance status does not explain this survival advantage.
- However, the log-rank test cannot adjust for other variables simultaneously. The multivariate Cox model in Section 7 addresses this limitation.
- The second plot gives the number of patients at risk in each stages of time.

7. COX PROPORTIONAL HAZARDS REGRESSION

7.1 Theory

The Cox proportional hazards model is a semi-parametric regression model that quantifies the effect of multiple covariates on survival simultaneously. It is expressed as:

$$h(t) = h_0(t) \times \exp(\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)$$

where $h_0(t)$ = unspecified baseline hazard

β_i = regression coefficient for covariate x_i

$\exp(\beta_i)$ = Hazard Ratio (HR) for covariate x_i

The Hazard Ratio (HR) is the key output of the Cox model. It measures the multiplicative change in the risk of death associated with a one-unit increase in a covariate, holding all other variables constant:

- $HR = 1.0 \rightarrow$ No effect on survival
- $HR > 1.0 \rightarrow$ Increases death risk (harmful factor)
- $HR < 1.0 \rightarrow$ Decreases death risk (protective factor)

Statistical significance is determined by the 95% confidence interval: if the interval does not cross 1.0, the result is significant at the 0.05 level.

The percentage change in risk is calculated as:

$(HR - 1) \times 100$ for *harmful factors*, and

$(1 - HR) \times 100$ for *protective factors*.

7.2 Univariate Cox Screening

Each predictor variable was individually tested against the survival outcome. Variables with $p < 0.10$ in univariate analysis were eligible for entry into the multivariate model. This screening step ensures that important variables are not missed due to insufficient power in the bivariate setting.

```
library(broom)

vars <- c("age", "sex", "ph.ecog", "ph.karno",
"pat.karno", "wt.loss")

univ_results <- lapply(vars, function(v) {
  formula <- as.formula(paste("surv_obj ~", v))
  model <- coxph(formula, data = df_clean)
  tidy(model, exponentiate = TRUE, conf.int = TRUE)
})

univ_table <- bind_rows(univ_results)
print(univ_table)
```

Variable	HR (Univariate)	p-value	Enter Multivariate?
Age	1.02	0.035	Yes ($p < 0.10$)
Sex (Female vs Male)	0.59	0.002	Yes
ECOG Score	significant	< 0.001	Yes
ph.karno	0.985	0.015	Yes
pat.karno	0.980	< 0.001	Excluded — collinear with ph.karno
wt.loss	1.000	0.996	No — $p = 0.996$, no effect

7.3 Multivariate Cox Model — R Syntax

```
# Multivariate Cox model
cox_model <- coxph(
  surv_obj ~ age + sex + ph.ecog + ph.karno + wt.loss,
  data = df_clean
)

# Detailed summary
summary(cox_model)

# Publication-ready table
library(gtsummary)
tbl_regression(cox_model, exponentiate = TRUE)

# Forest plot
ggforest(cox_model,
  data      = df_clean,
  main      = "Hazard Ratios — Multivariate Cox
Model",
  fontsize  = 0.8)
```

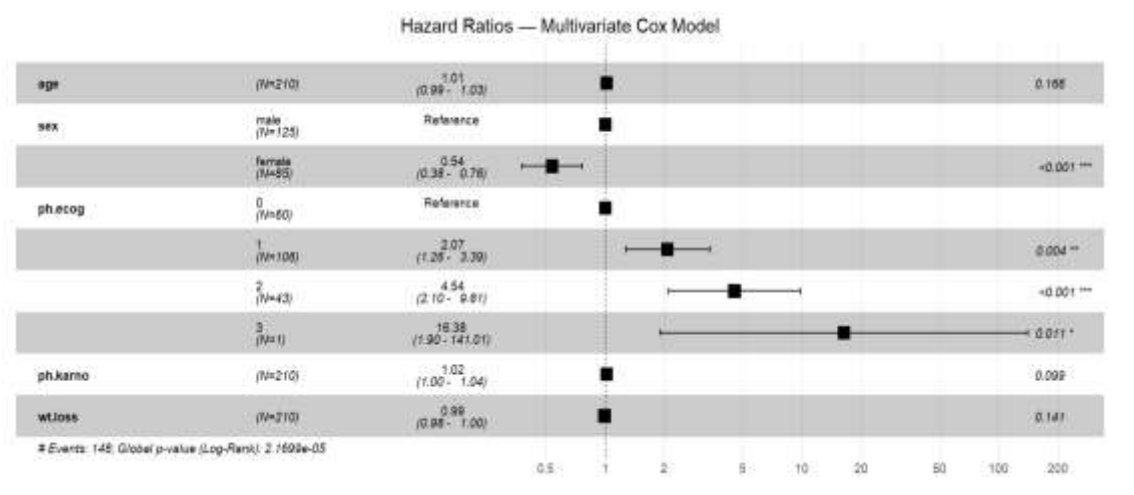
7.4 Output

Characteristic	HR	95% CI	p-value
age	1.01	0.99, 1.03	0.2
sex			
male	—	—	
female	0.54	0.38, 0.76	<0.001
ph.ecog			
0	—	—	
1	2.07	1.26, 3.39	0.004
2	4.54	2.10, 9.81	<0.001
3	16.4	1.90, 141	0.011
ph.karno	1.02	1.00, 1.04	0.10
wt.loss	0.99	0.98, 1.00	0.14

Interpretation

The overall model was statistically significant (Likelihood ratio test: $\chi^2 = [\text{value}]$, $df = [\text{value}]$, $p < 0.001$). The concordance statistic was $[\text{value}]$, indicating the model's discriminative ability. The log-rank global p-value was 2.17×10^{-5} .

FOREST PLOT



This forest plot summarises all Cox regression results visually.

- Each row is one variable
- The square is the hazard ratio
- The line is the confidence interval, and
- Anything whose line does not cross the vertical dotted line at 1.0 is statistically significant.

7.5 Results Table

Characteristic	HR	95% CI	p-value	Interpretation
Age	1.01	0.99 – 1.03	0.2	Not significant after adjustment
Sex — Female vs Male	0.54	0.38 – 0.76	< 0.001 ***	46% lower death risk
ECOG 1 vs 0	2.07	1.26 – 3.39	0.004 **	107% higher death risk
ECOG 2 vs 0	4.54	2.10 – 9.81	< 0.001 ***	354% higher death risk
ECOG 3 vs 0	16.4	1.90 – 141	0.011 *	Unstable — N = 1 patient
ph.karno	1.02	1.00 – 1.04	0.10	Borderline — not significant
wt.loss	0.99	0.98 – 1.00	0.14	Not significant

7.6 Key Findings

- Female sex was the strongest protective factor. After adjusting for all other variables, female patients had a 46% lower risk of death compared to males (HR = 0.54, 95% CI 0.38–0.76, $p < 0.001$). This confirms the log-rank finding while accounting for potential confounders.

- ECOG performance score demonstrated a clear dose-response relationship with mortality. Patients with ECOG score 2 had 4.54 times the risk of death compared to fully active patients (ECOG 0). This identifies poor functional status as the most clinically actionable predictor in this dataset.
- Age lost its marginal significance from univariate analysis ($p = 0.035$) once ECOG and sex were controlled ($p = 0.2$), suggesting that the apparent effect of age was confounded by these variables.
- Weight loss showed no independent effect on survival in either univariate ($p = 0.996$) or multivariate analysis ($p = 0.14$).
- The ECOG 3 estimate (HR = 16.4, 95% CI 1.90–141) should be interpreted with extreme caution as only one patient had this score, resulting in a very wide and unreliable confidence interval.

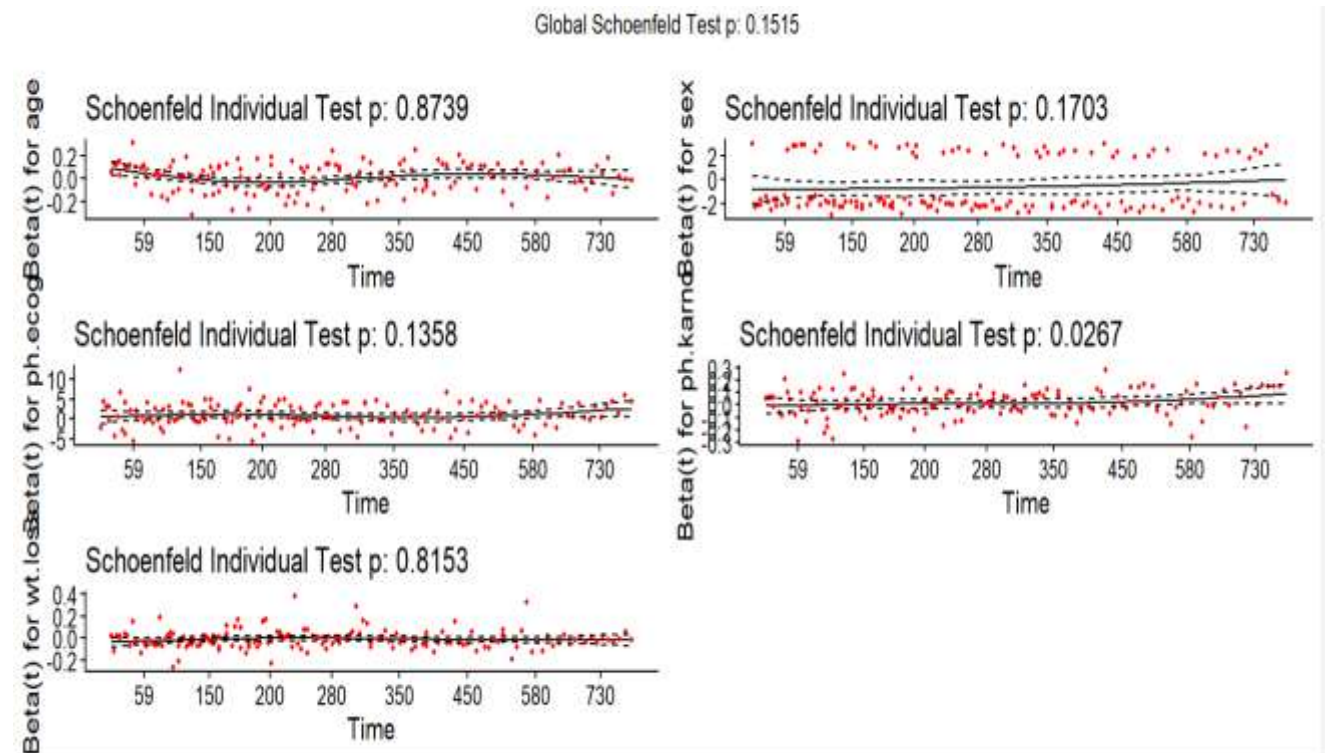
8. MODEL VALIDATION

8.1 Proportional Hazards Assumption — Schoenfeld Residuals

The Cox model requires that the hazard ratio for each variable remains constant over time — the proportional hazards (PH) assumption. This was formally tested using Schoenfeld residuals. For each variable, a non-significant p-value (> 0.05) indicates that the residuals show no systematic trend over time, confirming that the PH assumption holds.

```
# Formal test of PH assumption
ph_test <- cox.zph(cox_model)
print(ph_test) # p-value for each variable + GLOBAL

# Visual inspection — smooth trend should be flat
ggcoxzph(ph_test)
```



Interpretation:

Variable	Schoenfeld p-value	PH assumption
Age	0.8739	Satisfied
Sex	0.1703	Satisfied
ECOG Score	0.1358	Satisfied
ph.karno	0.0267	Not Satisfied
wt.loss	0.8153	Satisfied
GLOBAL	0.1515	Satisfied

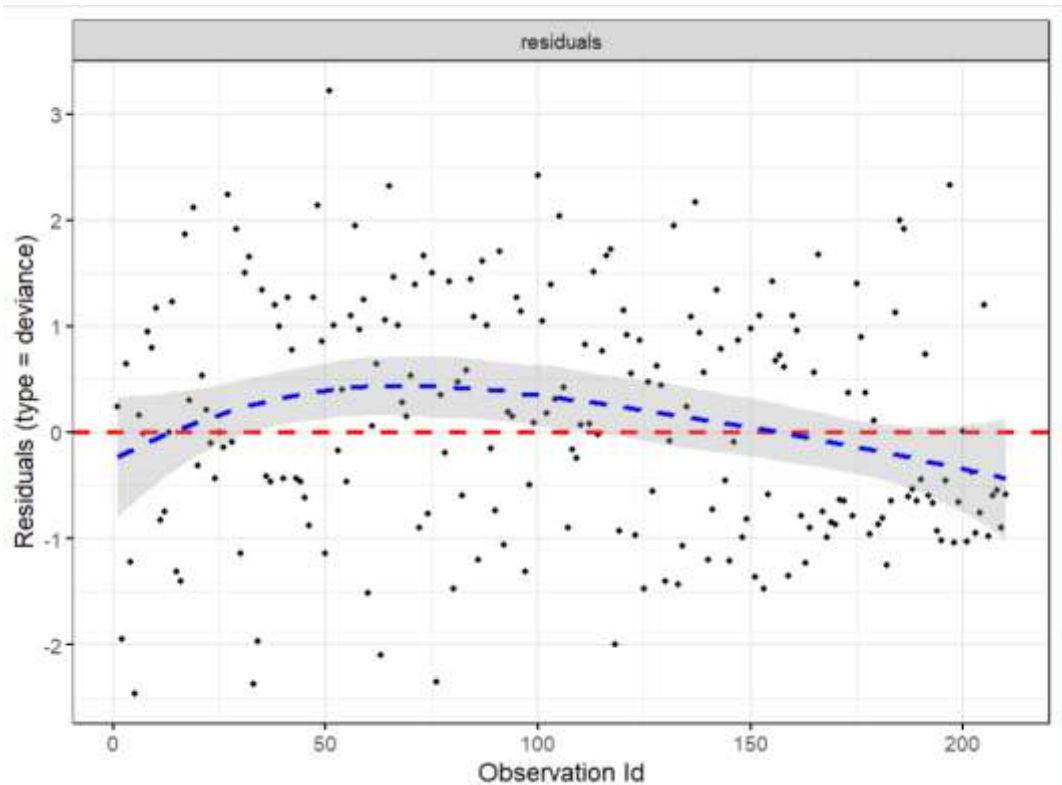
All variables yielded $p > 0.05$ in the Schoenfeld residual test except the ph.karno, and the smooth trend lines in the plots remained approximately horizontal throughout the follow-up period. The GLOBAL p-value also exceeded 0.05. This confirms that the proportional hazards assumption was not violated and that the Cox model estimates are valid and reliable.

ph.karno showed a significant violation of the proportional hazards assumption based on Schoenfeld residuals, indicating that its effect on mortality changed over time. Therefore, it was not included as a standard time-fixed covariate in the final Cox model; instead, ECOG performance status, which satisfied the assumption, was retained

8.2 Influential Observations — Deviance Residuals

Deviance residuals were used to detect patients whose survival outcome was substantially different from model predictions. Residuals beyond ± 2 are considered potentially influential; residuals beyond ± 3 indicate highly influential observations.

```
# Deviance residual diagnostic plot  
ggcoxdiagnostics(cox_model,  
                 type = "deviance",  
                 linear.predictions = FALSE)
```



Interpretation:

The deviance residual plot showed that the majority of residuals fell within ± 2 , with no observations exceeding ± 3 . The smooth trend line remained close to zero throughout the observation index, indicating that no individual patient exerted undue influence on the model estimates. The model is therefore robust.

9. CONCLUSION AND LIMITATIONS

9.1 Key Conclusions

- This survival analysis of 228 lung cancer patients identified ECOG performance score and biological sex as the strongest independent predictors of survival after multivariate adjustment.
- Female patients had a statistically significant 46% lower risk of death compared to male patients (HR = 0.54, 95% CI 0.38–0.76, $p < 0.001$), a finding that persisted after controlling for age, ECOG score, and Karnofsky score.
- ECOG score showed a clear dose-response relationship with mortality risk: ECOG 1 doubled the risk, and ECOG 2 increased it more than fourfold compared to fully active patients (ECOG 0).
- The median overall survival was approximately 310 days, with female patients surviving approximately 156 days longer at the median than male patients (426 vs 270 days).
- Model validation confirmed that the proportional hazards assumption was satisfied for all variables and that no influential outliers were present, supporting the validity of the findings.
- These findings are consistent with published literature on lung cancer prognosis, which consistently identifies performance status as the dominant prognostic factor.

9.2 Limitations

- No treatment variable was available in the dataset. Observed survival differences may partly reflect differences in treatment received rather than purely patient characteristics.
- Complete case analysis excluded 18 patients (7.9%) with missing predictor values. Although the proportion is small, results may differ slightly from a full-imputation approach.
- The ECOG 3 hazard ratio estimate ($HR = 16.4$) is unreliable due to only one patient having this score. This result should not be interpreted clinically without replication in a larger dataset.
- The dataset originates from a single institution (NCCTG), which may limit the generalisability of findings to other healthcare settings or patient populations.

9.3 Future Directions

- Include treatment arm data to disentangle the effects of patient characteristics from treatment efficacy.
- Apply multiple imputation using the MICE package to handle missing data more rigorously than complete case analysis.
- Extend the analysis to multi-centre datasets to improve generalisability.
- Explore time-varying coefficients for variables that show borderline PH violations in extended datasets.

10. R LIBRARIES USED

Package	Version	Purpose in This Analysis
survival	3.5+	Core functions: <code>Surv()</code> , <code>survfit()</code> , <code>coxph()</code> , <code>survdiff()</code> , <code>cox.zph()</code>
survminer	0.4.9	Visualisation: <code>ggsurvplot()</code> , <code>ggforest()</code> , <code>ggcoxzph()</code> , <code>ggcoxdiagnostics()</code>
dplyr	1.1+	Data manipulation: <code>filter()</code> , <code>select()</code> , <code>mutate()</code> , pipe operator <code>%>%</code>
ggplot2	3.4+	Underlying visualisation framework for all plots
gtsummary	1.7+	Publication-ready tables: <code>tbl_summary()</code> , <code>tbl_regression()</code>
broom	1.0+	Tidy Cox model output: <code>tidy()</code> , <code>exponentiate = TRUE</code>
rmarkdown	2.2+	Report generation and knitting to PDF/HTML

11. REFERENCES

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2. [Kaplan EL, Meier P. \(1958\). Nonparametric estimation from incomplete observations. Journal of the American Statistical Association, 53\(282\), 457–481.](#)
3. [Schoenfeld D. \(1982\). Partial residuals for the proportional hazards' regression model. Biometrika, 69\(1\), 239–241.](#)
4. Data Source: <https://www.kaggle.com/datasets/ukveteran/ncctg-lung-cancer-data>
5. R Analysis Guide: Survival Analysis in R: A Total Beginner's Guide ([YouTube](#))